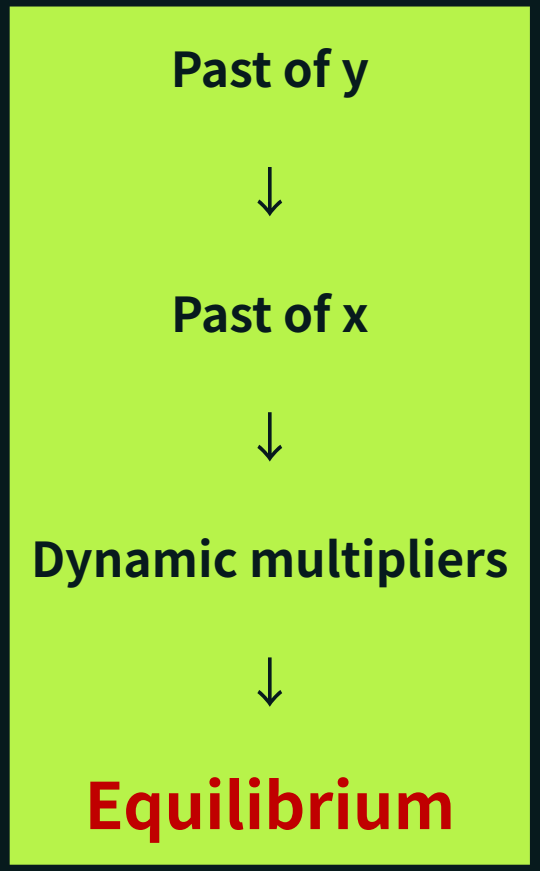
The ADL Model
Lagged Effects

Explaining y with its own past and the past of x



We read the ADL model in four steps

01	Structure and estimation	**AR part + DL part** Predictive vs structural · special cases · OLS
02	Dynamic multipliers	**How the effect spreads** Impact · dynamic · long-run multipliers
03	Error-correction form (ECM)	**Same model, rewritten** Long-run equilibrium · gap · adjustment speed
04	Finance cases and labs	**Pass-through and forecasting** FX & rate pass-through · oil · Claude Code

Structure → path → equilibrium → cases

Economic effects arrive with lags and persistence

Three reasons we need ADL (Autoregressive Distributed Lag) models

LAG

Effects arrive late

Rate cut → lagged

PERSISTENCE

Own past of y

Else x is overstated

FORECAST

Forecast from the past

No need for current x

AUTOREGRESSIVE

AR part

Past y as regressors

DISTRIBUTED LAG

DL part

Past x as regressors

LAG ORDER

p and r

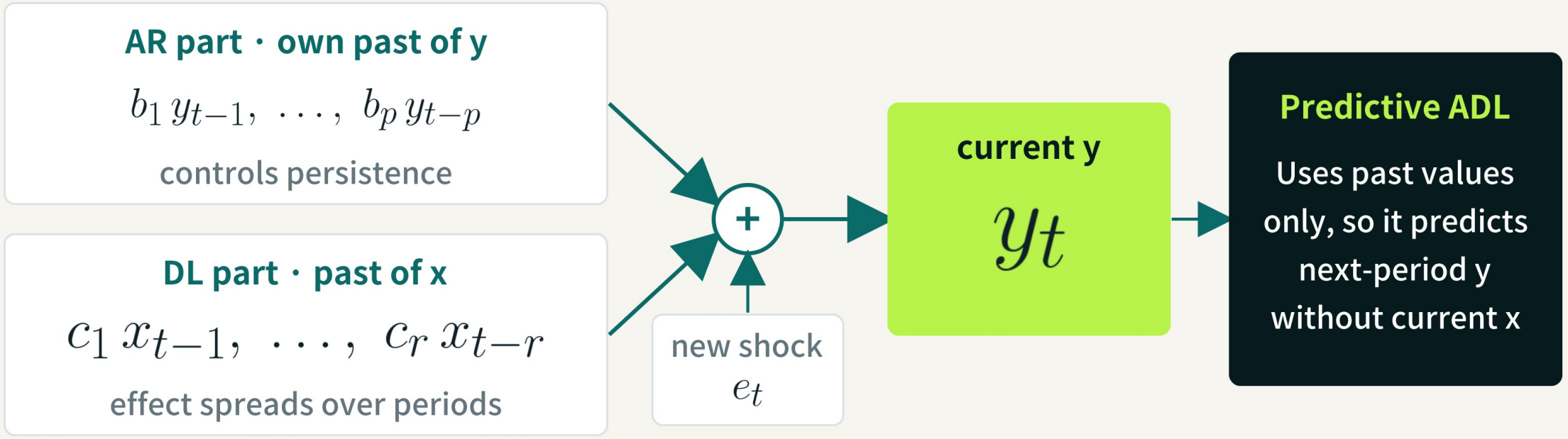
Max lags of y and x

Only after controlling persistence does the pure effect of x show.

ADL = one equation explaining current y with past y and past x

ADL combines the AR part and the DL part in one equation

Past y and past x add up; what remains unexplained is the new shock



It captures dynamics where the effect of x spreads over several periods.
Using only past values, the predictive form is a forecasting model as is

Including current x separates the predictive and structural forms

ADL(p, r) uses p lags of y and r lags of x

Predictive ADL(p, r) — the form used for Granger causality tests

$$y_t = a + b_1 y_{t-1} + \cdots + b_p y_{t-p} + c_1 x_{t-1} + \cdots + c_r x_{t-r} + e_t$$

Structural ADL — adds the contemporaneous effect; requires exogenous x

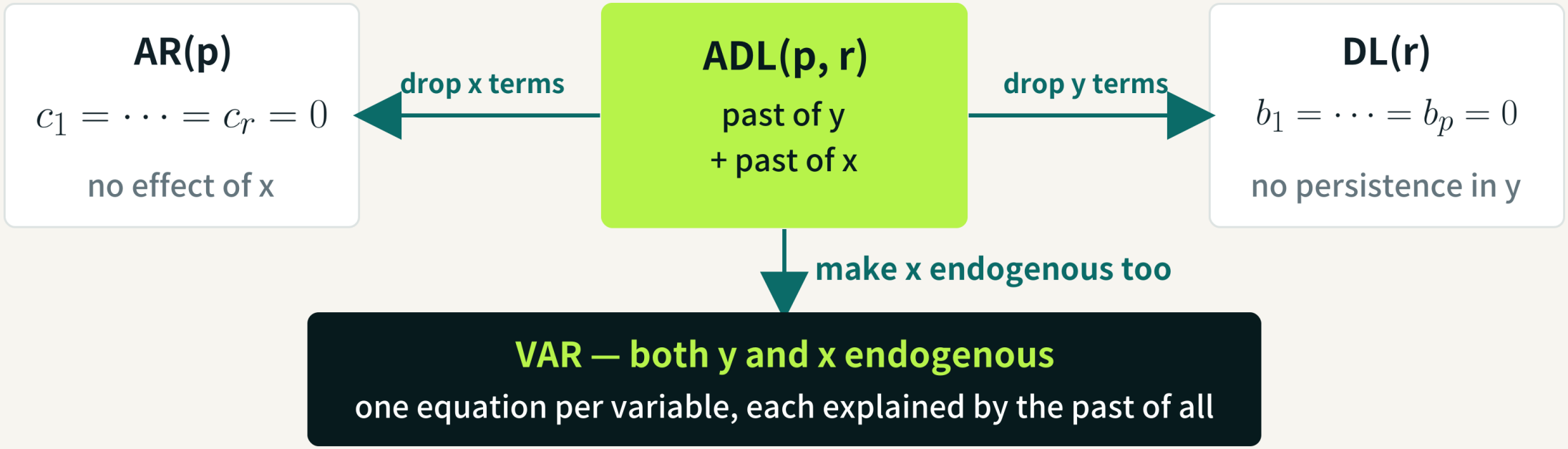
$$y_t = a + b_1 y_{t-1} + \cdots + b_p y_{t-p} + \boxed{c_0 x_t} + c_1 x_{t-1} + \cdots + c_r x_{t-r} + e_t$$

The predictive form computes next-period y without current x.

The boxed term is the immediate effect of same-period x on y

Zero restrictions give AR and DL; extending it gives a VAR

Placing familiar models around the ADL



AR and DL are restricted special cases of ADL.

If y also drives x, move from a single equation to a VAR

Estimate by OLS after checking three conditions

Minimize the sum of squared errors — the first observations, up to the longest lag, drop out

CONDITION 1

Stationarity

Else spurious fit

CONDITION 2

Uncorrelated errors

Else OLS fails

CONDITION 3

Exogenous past x

By design if predictive

Lag selection Minimize AIC/BIC | Ljung–Box on residuals | Check white noise

With past y among the regressors, autocorrelated errors bias the estimates.

Residual tests are mandatory, not optional

Dynamic multipliers trace the path of y after a shock to x

Dynamic multipliers — how y responds over time when x rises once by one unit

Definition

$$d_h = \frac{\partial\, y_{t+h}}{\partial\, x_t}, \qquad k = d_0 + d_1 + d_2 + \cdots$$

IMPACT MULTIPLIER

Impact

First response

DYNAMIC MULTIPLIER

Dynamic

Response after h

LONG-RUN MULTIPLIER

Long-run

Lasting-rise effect

The impact multiplier is the start, dynamic ones the path, the long-run one the destination.

The long-run multiplier is the sum of all dynamic multipliers

The long-run multiplier scales the x coefficients by y's persistence

Sum of x coefficients over one minus the sum of y coefficients; c0 is zero if predictive

Long-run multiplier k — equilibrium slope

$$k = \frac{c_0 + c_1 + \cdots + c_r}{1 - (b_1 + \cdots + b_p)}$$

Stability condition

$$b_1 + \cdots + b_p < 1$$

Why it matters

- Denominator > 0: effect converges
- k grows as persistence → 1

The numerator is the direct push of x; the denominator, how persistence amplifies it.

Near-unit persistence turns small coefficients into large long-run effects

The example peaks in period two and converges to 1.0

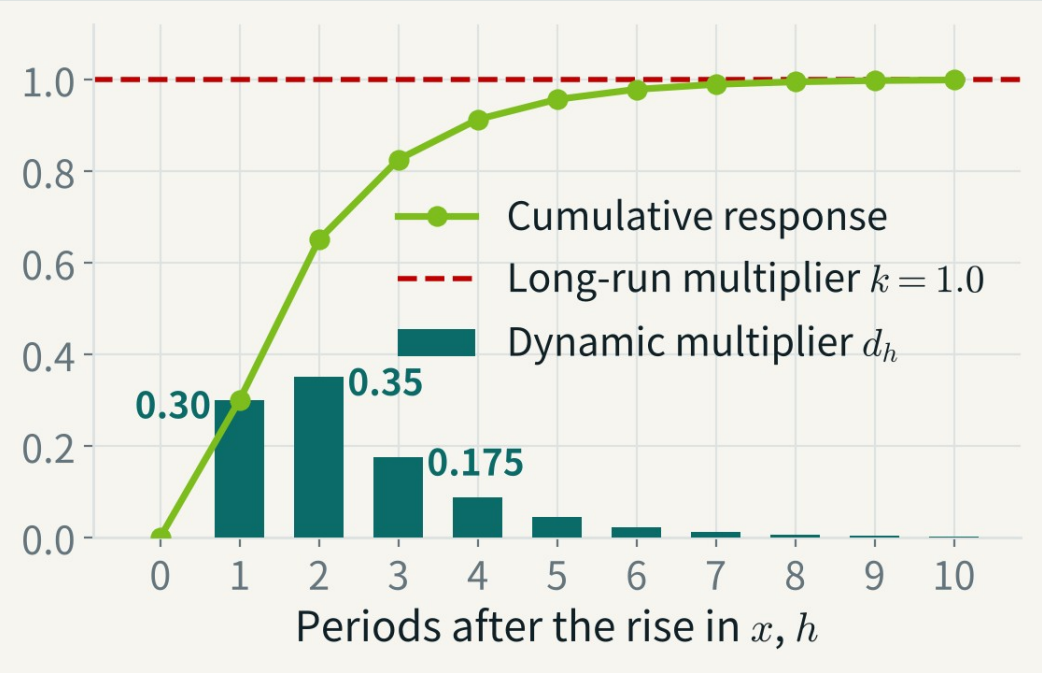
Dynamic multipliers and cumulative response after a one-unit rise in x — calculations are in the workbook

Example model

$$y_t = 1 + 0.5\,y_{t-1} + 0.3\,x_{t-1} + 0.2\,x_{t-2} + e_t$$

- Impact 0.30 (h = 1)
- Peak 0.35 (h = 2)
- Then halves

Cumulative → 1.0



Being predictive, the same-period response is 0; the effect starts next period.

The cumulative response matches the long-run multiplier formula exactly

ADL and ECM are the same model written differently

Rewriting ADL(1,1) in changes separates the short-run response from gap adjustment

ADL form · levels

$$y_t = a + b_1 y_{t-1} + c_0 x_t + c_1 x_{t-1} + e_t$$

regresses y on x in levels as is

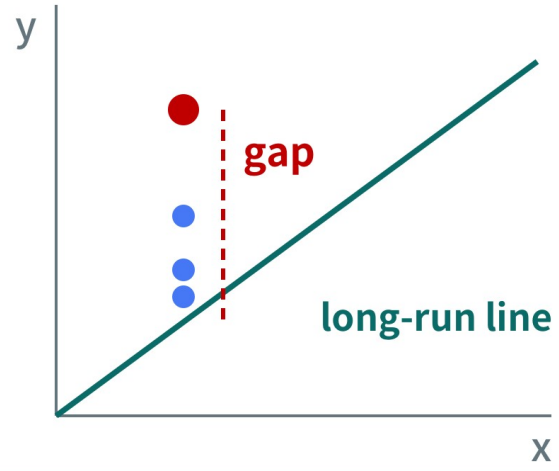
same model

ECM form · changes

$$\Delta y_t = a + \underbrace{c_0 \Delta x_t}_{\text{short run}} + a_y \underbrace{[y_{t-1} - k\, x_{t-1}]}_{\text{gap}} + e_t$$

short-run response + closing part of last period's gap

The gap shrinks each period



Written in levels it is an ADL; in changes and the gap, an ECM.

The ECM shows equilibrium and adjustment speed as coefficients

The ECM closes part of last period's gap every period

Error-correction model — starting from ADL(1,1)

Error-correction form

$$\Delta y_t = a + c_0\,\Delta x_t + a_y\Big[\,y_{t-1} - k\,x_{t-1}\Big] + e_t$$

Link to ADL coefficients

$$a_y = -(1 - b_1), \qquad k = \frac{c_0 + c_1}{1 - b_1}$$

LONG-RUN MULTIPLIER

Slope of the line

In the long run y = k·x

IN BRACKETS

Last period's gap

Off-equilibrium size

ADJUSTMENT SPEED

Negative to revert

Share of the gap closed

With persistence 0.5 the adjustment speed is −0.5: the gap halves each period.

This structure is the starting point for cointegration and the VECM

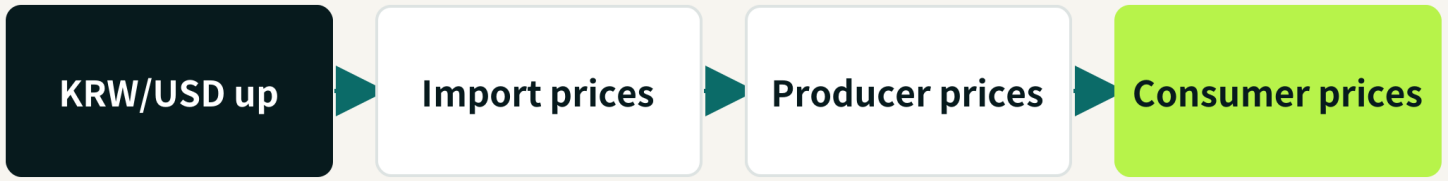
Exchange-rate pass-through is gradual and usually incomplete

Import price inflation explained by its own lags and lags of KRW/USD changes

Episode · 2008 & 2022 spikes

- KRW/USD surges
- Import prices jump
- CPI follows later

Long-run k < 1



Passed through stage by stage, with lags

Impact multiplier
= short-run pass-through

Long-run multiplier
= long-run pass-through (< 1)

Why incomplete: margins absorb · non-USD invoicing · FX hedging

Read short-run pass-through from the impact multiplier, long-run from the long-run one.

We can sketch the price path during an FX spike in advance

Policy-rate pass-through is judged by k and adjustment speed

An ADL explaining lending rates with their own lags and lags of the policy rate

What to watch

Long-run k near 1

- 1 means full pass-through
- Check adjustment speed too

Bank competition & funding

Episode · rate cycles

Up fast, down slow

- Hikes passed on at once
- Cuts passed on slowly

Asymmetric pass-through

Estimating hiking and cutting phases separately reveals asymmetric pass-through.

Supervisors and banks watch both the long-run multiplier and adjustment speed

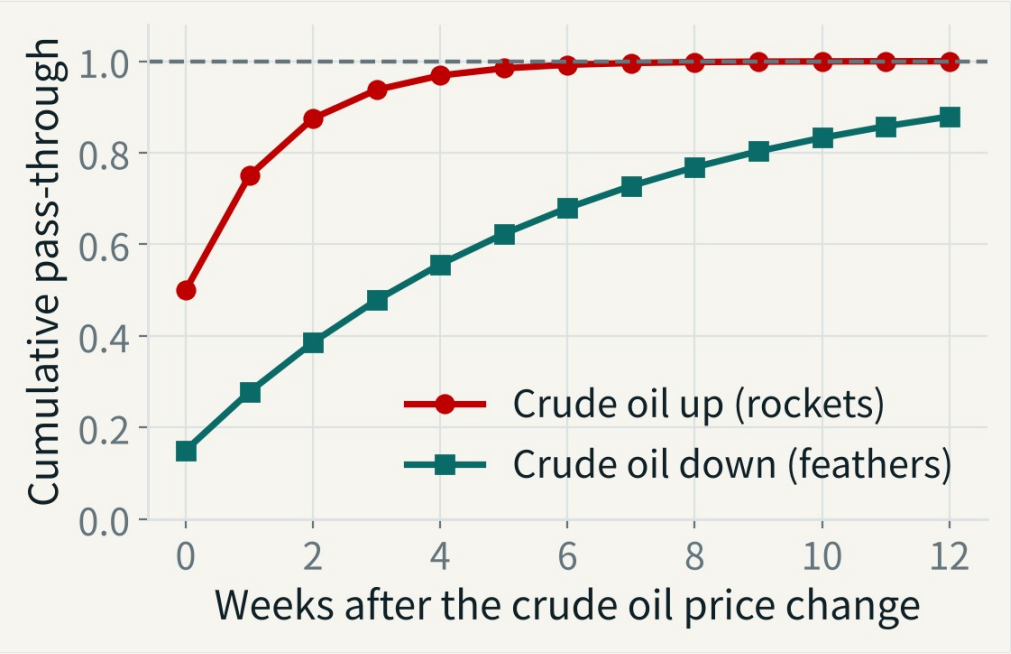
Gasoline prices rise like rockets and fall like feathers

Rockets and feathers (Bacon, 1991) — cumulative pass-through when crude rises vs falls

Episode · slow price cuts

- Crude already fell
- Pump prices lag
- Rises pass at once

Compare up/down multipliers



※ Simulated data — illustrative paths showing the shape of the asymmetry

Test the asymmetry with a nonlinear ADL (NARDL) splitting x into rises and falls

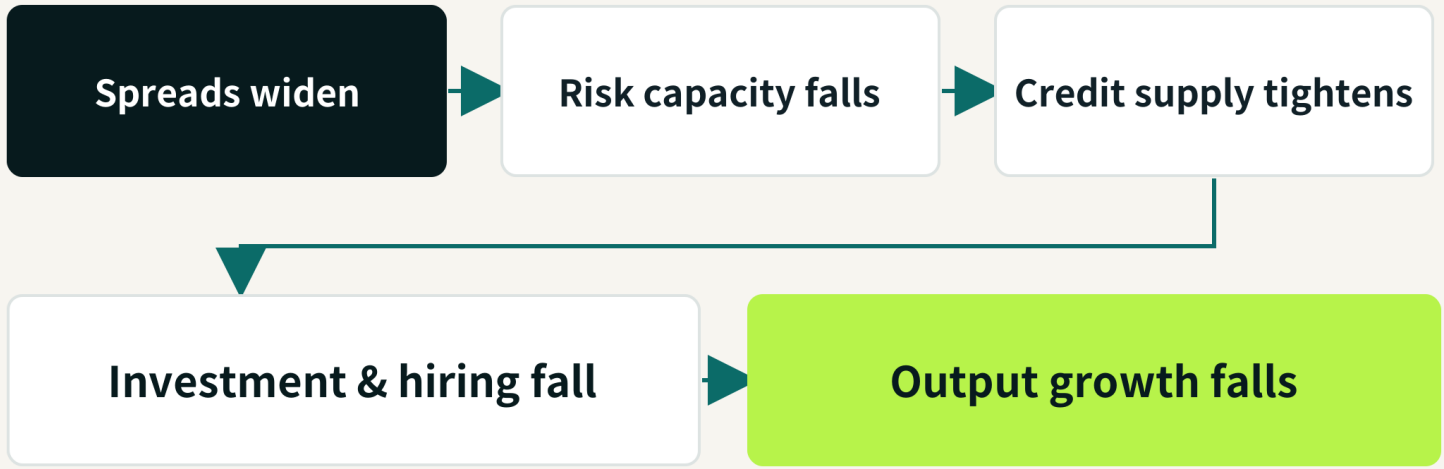
Credit spreads signal economic contractions in advance

Forecasting IP growth with its own lags and lags of corporate bond spreads (Gilchrist & Zakrajsek, 2012)

Episode · 2008 crisis

- Spreads spike first
- Output falls later

Better OOS forecasts



Spreads lead output indicators → forecast y with the past of x

Among spreads, the excess bond premium (EBP) leads most strongly.

Evaluate a predictive ADL by its out-of-sample forecast errors

A practitioner report states three things together

A checklist that lets readers of pass-through and forecasting work judge at a glance

CHECK 1

Long-run + adjustment

- ECM form reads easily
- Test against 1

CHECK 2

Asymmetry via NARDL

- Split rises and falls
- Overlay responses

CHECK 3

Forecast out of sample

- Not in-sample fit
- Out-of-sample forecast errors

Reporting a single number misses the speed and direction of pass-through.

Show the path, the destination and the asymmetry on one page

Lab ①: Claude Code estimates oil-to-gasoline pass-through

Asymmetry included — FRED DCOILWTICO (WTI, weekly avg.) and GASREGW (retail gasoline) from 2000

Stationarity diff if needed → **Pick lags by BIC estimate ADL** → **Multipliers 12-week plot** → **NARDL up vs down** → **md report open it**

OUTPUT

adl_rockets_feathers.md

Long-run pass-through + asymmetry plot

LAST LINE OF PROMPT

Fix errors on its own

Instructed to run to the end

Students only paste the prompt and check the results

Lab ②: Claude Code estimates KRW/USD exchange-rate pass-through

Download FRED DEXKOUS (KRW/USD, monthly average) and a Korean import price index from 2005

Find data or substitute → **Monthly changes estimate ADL** → **Multipliers test vs 1** → **Mark spikes 2008 & 2022** → **5-line summary save md, open**

OUTPUT

adl_pass_through.md

5-line summary for practitioners

DECISION POINT

Explain substitutes first

Reasons before I choose

Humans decide only where judgment is needed; Claude Code runs the rest

ADL holds the short-run path and long-run equilibrium together

Two equations and four messages to remember from this section

Long-run multiplier

$$k = \frac{c_0 + c_1 + \cdots + c_r}{1 - (b_1 + \cdots + b_p)}$$

Error-correction form

$$\Delta y_t = a + c_0 \, \Delta x_t + a_y \left[y_{t-1} - k \, x_{t-1} \right] + e_t$$

Key messages

- Control persistence + lags
- Dynamic = path, k = destination
- ECM: gap + adjustment speed
- NARDL; out-of-sample tests

Next: cointegration and the VECM extend the gap to several variables.

ADL is the standard tool for pass-through analysis

EFFECTS ARRIVE WITH LAGS.

MULTIPLIERS MAP THE PATH.

THE GAP CLOSES OVER TIME.

Read effects that spread over time
as a path and an equilibrium.